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To cite this article: Ellis Adjei Adams *et al* 2023 *Environ. Res. Lett.* **18** 123001

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RECEIVED
11 March 2023

REVISED
21 October 2023

ACCEPTED FOR PUBLICATION
25 October 2023

PUBLISHED
3 November 2023

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TOPICAL REVIEW

Farmer–herder conflicts in sub-Saharan Africa: drivers, impacts, and resolution and peacebuilding strategies

Ellis Adjei Adams[1,2,4,*], **Audrey Thill**[2], **Elias Danyi Kuusaana**[3] **and Anna Mittag**[4]

[1] Keough School of Global Affairs, University of Notre Dame, Notre Dame, IN 46556, United States of America
[2] Kroc Institute for International Peace Studies, University of Notre Dame, Notre Dame, IN 46556, United States of America
[3] Department of Land Management, SD Dombo University of Business and Integrated Development Studies, Post Office Box UPW 3, Wa, Upper West Region, Ghana
[4] Kellogg Institute for International Studies, 1130 Jenkins Hall, Notre Dame, IN 46556, United States of America
[*] Author to whom any correspondence should be addressed.

E-mail: eadams7@nd.edu

Keywords: farmers–herder conflict, resolution, conflict management, peacebuilding, environmental security

Abstract

Resource scarcity and insecurity due to climate change, coupled with ethnic, religious, and identity politics, have strained peaceful coexistence between farmers and herders in West and Central Africa over the last few decades. Violent farmer–herder clashes now pose a significant threat to security and stability in the Sahelian and savannah dryland regions. Despite the large and increasing literature on these conflicts in recent decades, no comprehensive reviews or syntheses of knowledge on these conflicts have been published. We review empirical studies on farmer–herder conflicts in sub-Saharan Africa published from 2000 to 2021, focusing holistically on drivers, impacts, and the opportunities and limits of the main resolution and peacebuilding strategies different actors use to manage the conflicts. Most of the studies focused on Nigeria, Ghana, Tanzania, and Mali and mainly covered conflict dynamics, political ecology, agriculture, climate change, conflict management, governance, land tenure, and peacebuilding. The majority were cross-sectional and relied on qualitative data and mixed methods. The reviewed literature shows that main drivers of farmer–herder conflicts are climate-induced environmental factors, land tenure insecurity, resource competition and intergroup conflicts and prejudices. Conflict control and prevention strategies fell along a spectrum of aggressive, through passive, to constructive tactics. The main conflict management strategies used included third-party intervention (mediation and arbitration), litigation, and direct negotiations for damage compensation. Key actors involved in peacebuilding efforts included farmers and herders themselves, community leaders, community-based organizations, police and courts, and informal community groups. Our findings and discussion advance broader understanding of farmer–herder conflicts, establish the opportunities and limits of various resolution approaches, and highlight gaps and directions for relevant future research.

1. Introduction and background

Farmers and herders have historically coexisted and shared resources in sub-Saharan Africa. However, resource scarcity and insecurity due to climate change, coupled with ethnic, religious, and identity politics, have strained peaceful coexistence over the last few decades. Now violent farmer–herder clashes pose a significant threat to security and stability in the Sahelian and savannah dryland regions of West and Central Africa. Across the Sahel, the number of casualties is growing, and there has been a significant loss of property. In June 2018, the British Broadcasting Corporation reported that 86 people had died, and 50 houses had been burnt in Nigeria's plateau state due to violent clashes between farmers and *Fulani herders* (BBC News 2018). Amnesty International also reported that more than 3641 lives had been lost in Nigeria because of farmer–herder conflicts between 2016 and 2018 (Amnesty International 2018). ACLED reported that this violence marked a five-year high in recorded violent

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incidents between Fulani herders and farming communities in Nigeria (ACLED 2018). In Ghana, some studies have estimated that over 40 farmers have died from farmer–herder clashes in the Agogo area, while *Konkomba farmers* killed over 40 Fulani herders in the Gushiegu area (Bukari *et al* 2018). In addition to the loss of human and animal lives, farmlands and shared water resources have also been damaged by these conflicts. In numerous incidents, violence between farmers and herders has been triggered by the destruction of farmlands by cattle (Alhassan 2017, Yakubu *et al* 2020) or by the pollution of shared water resources (Tade and Yikwabs 2019, Otu *et al* 2020).

These violent clashes are becoming more frequent and intense because of climate-induced drought, land scarcity and tenure insecurity, agricultural expansion, and commercial land grabbing (Kuusaana and Bukari 2015, Akov 2017). Drought, desertification, decreasing grazing land, and declining water resources are forcing herders to move into new territories near farmlands and agricultural communities, triggering fresh clashes and reigniting old ones. While intensive competition over declining resources, especially land and water, is at the core of these sometimes-violent conflicts, territory, religion, language, ethnicity, and land rights also play important roles (Bukari and Schareika 2015, Olaniyan *et al* 2015).

An abundant literature on farmer–herder conflict in sub-Saharan Africa has documented a number of these drivers of farmer–herder conflicts. Many are centered around cultural differences (Kuusaana and Bukari 2015), stereotypes and biases (Bukari *et al* 2018), land tenure insecurity, and climate-induced environmental factors. Some researchers have examined these environmental factors from the perspective of environmental determinism, with the core argument being that the struggle over land-based resources is a major driver of these conflicts (Adams *et al* 2019, Walwa 2020). For instance, as land, grass, and water become scarce, farmers and herders compete among themselves to outsmart each other (Dimelu *et al* 2016, Mahmood and Azuaga 2020). Large-scale land acquisitions have also restricted land use for both farmers and herders, triggering conflicts among farmers, herders, and investors (Bukari *et al* 2018). Other studies have highlighted sociocultural factors as the primary drivers of farmer–herder conflicts, for example, constructed identities and superiorities (Bukari *et al* 2018). Fulani herders are identified as aliens, foreigners, strangers, non-natives, non-citizens, thieves, rapists, or murderers by farmers (Kuusaana and Bukari 2015). Other authors have found that ethnic and religious discrimination underlie farmer–herder conflicts and influence how smallholders respond to herder encroachment on their land (Bukari and Schareika 2015, Olaniyan *et al* 2015).

Despite this abundant literature on farmer–herder conflicts in sub-Saharan Africa, there are scarcely any comprehensive reviews and syntheses of the nuances and dynamics of these conflicts and how they are resolved. To our knowledge, the only existing comprehensive review of farmer herder cases relied on a database of actual conflict cases rather than peer reviewed literature on the farmer–herder phenomenon and was published as a report (Krätli and Toulmin 2020). In the present study we set out to review the literature on farmer–herder conflicts in sub-Saharan Africa over the last two decades to identify the historical and emerging drivers of the conflicts; their impacts on livelihoods, agriculture, and food security across the region; and the opportunities and limitations of the various resolution and peacebuilding strategies applied in various conflict contexts. In the next section, we describe the methods used for the review search and analysis before sharing our results and discussion.

2. Review methods

2.1. Literature search, screening, and selection strategy

Given the breadth and heterogeneity of literature available on the topic of farmer–herder conflicts, we opted for a narrative review to enable a more comprehensive coverage. A narrative review is a comprehensive analysis of the state of knowledge on a given topic. Opting for a narrative review allowed us to search for literature on different themes related to the subject matter, rather than use a systematic review which is ill suited because it is more useful when there is a narrow and more specific research focus or question and in cases where quantitative studies dominate the literature (Collins and Fauser 2005, Berrang-Ford *et al* 2015). Following the approach specified for narrative reviews by Berrang-Ford *et al* (2015), we applied a systematic procedure for our document selection and inclusion.

For this study, we reviewed peer-reviewed reports of original research published between 2000 and 2021 in English-language journals. Although grey literature such as policy and NGO documents have proliferated on this topic, our review focuses on peer-reviewed articles which we assume to have more systematic and rigorous methodological approaches to studying this issue. Prior to the identification phase, we selected key search terms based on prior knowledge of the subject, discussions among the research team, and a preliminary literature search. These included *farmer–herder conflicts, herder–farmer conflicts, farmer–pastoralist conflicts, farmer–herder violence, armed farmer–herder conflicts,* and *farmer–herder clashes*. Figure 1 presents an overview of the literature search strategy. We identified articles mainly through a search of the Google Scholar database for articles with titles that included our key search terms. Next, we conducted a keyword search of the Science Direct database and downloaded all related

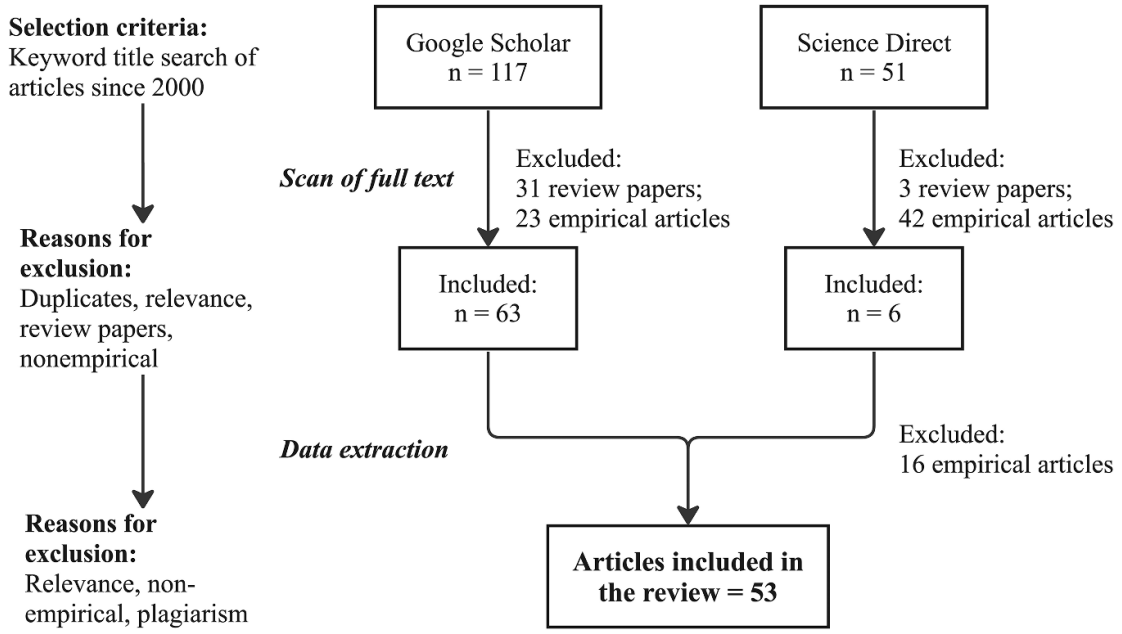

Figure 1. Flow chart of literature selection process.

articles. We excluded articles that we had previously downloaded from Google Scholar. The initial pool of downloads included 117 articles from Google Scholar and 51 articles from Science Direct.

During the second screening phase, we scanned the introduction, methods, results, and discussion sections of each article and excluded those that were duplicates, review papers, opinion or concept papers, grey literature, studies that did not directly involve pastoralists or farmers, and studies that focused on regions rather than countries ($n = 99$ excluded). The included studies were empirical, peer reviewed, rigorous, and highlighted country-specific contexts. We then read the full text of each article, excluding seven more, several of which did not meet the basic quality requirements or had significant grammatical and punctuation errors, potentially indicating that they were not critically peer reviewed. Finally, as we began to extract and summarize the information from the articles, we excluded nine additional articles that did not meet our criteria, were duplicates, or in one case, included plagiarized material. At the end of this screening process, 53 articles remained for inclusion in the review.

2.2. Data extraction and coding

In the next phase, we developed a table with key thematic categories using Microsoft Excel. These categories, which are listed in table 1, guided data extraction. Two of the authors independently reviewed and coded an initial set of articles, convened to discuss and resolve differences and disagreements, edited the spreadsheet for clarity, and divided and coded the rest of the articles. All the data relevant to these categories were compiled in a spreadsheet. When an article contained a large amount of data in one category, we included the data most salient to this review. For example, if the informants in an article identified 20 causes of farmer–herder conflicts, we included only the top three to ensure consistency with the other articles.

We used a thematic approach to coding and data extraction to analyze patterns and synthesize data, following Shaffril *et al* (2021). We carefully paid attention to recurrent themes from the articles and wrote summaries on individual word documents for further analysis. In our thematic analysis, we intentionally integrated both inductive and deductive approaches, drawing insights from Proudfoot (2023)'s inductive/deductive hybrid thematic analysis. We integrated pre-determined themes based on our knowledge of the literature with emergent themes from reading the articles to allow for a rich and expansive set of themes. After extracting and cleaning the data, we analyzed each thematic category separately to identify the patterns and subthemes within them. We created an Excel spreadsheet for each category and its subthemes. We then coded each line of the spreadsheet and counted the number of articles aligned with each subtheme. For example, we identified the main types of conflict-resolution actors mentioned in the articles, coded each line of the spreadsheet that had each type of actor, and then counted the number of articles that mentioned each actor. We then converted some of the data into bar graphs and charts, some of which are presented in the results. We then used

Table 1. Thematic categories used for data extraction.

Category	Example from Bukari *et al* (2018)
Publication year	2018
Location of study	Ghana
Name of journal	*African Studies Review*
Main theme(s)	Conflict; everyday peace
Research objectives or questions	To study cooperative relationships between farmers and herders in Ghana. Are conflicts as frequent as implied by public discourses?
Methodology	Purely qualitative
Type of data analyzed	Focus group interviews; participant observations; ethnographic interviews with farmers, Fulani, cattle owners, local officials, and community leaders and members
Analytical method	Comparative analysis; theoretical analysis
Theoretical framework	Everyday peace
Summary of article's findings	Farmers and herders in Ghana are 'cultural neighbors,' meaning they inhabit the same communities despite some ethnic, social, identity, and cultural difference…
Major cause(s) of farmer–herder conflicts	Disputes over land increased farmer–herder hostilities; disagreements about the level of compensation Fulani should pay farmers
Major impact(s) of farmer–herder conflicts	Death, injuries, destruction of property (homes, crops, cattle).
Conflict-prevention or -control strategies	Trust building through everyday mutually beneficial social and economic interactions
Conflict-resolution actors/stakeholders	1-Farmers and herders, 2-Unit committee members, 3-Village leaders, 4-Police, 5-State forces
Responsibilities of stakeholders	1-Negotiation, 2-Mediation, 3-Arbitration, 4-Peacekeeping
Conflict-resolution approach(es)	Compensation, mediation, arbitration, peacekeeping
Research gaps/future work	None identified/proposed.
Policy recommendations	More bottom-up approaches to peacebuilding between farmers and herders

summary statistics to write annotated summary paragraphs for each thematic category. For some categories, we extracted additional data to provide supporting examples from the text. We then read these summary paragraphs and identified themes and points of analysis that needed more detail. Finally, we synthesized and built on these paragraphs to produce this review.

3. Findings

We structured the findings of our review into six core domains: objectives, and geographical scope; main themes; methods and analytical techniques; theoretical and conceptual frameworks; the major causes of farmer–herder conflicts; and strategies for conflict prevention and peacebuilding.

3.1. Objectives and geographical scope

The primary objective of 68% of the reviewed articles was to address the drivers and dynamics of farmer–herder conflicts. Farmer–herder conflicts are driven by multiple factors, and the dynamics characterizing these conflicts are dependent on the contexts and actors involved. For example, some articles focused on drivers such as cultural biases (Mahmood and Azuaga 2020) and land tenure insecurity (Benjaminsen *et al* 2009, Kuusaana and Bukari 2015, Mwikali and Wafula 2019, Kugbega and Aboagye 2021). Others primarily analyzed environmental drivers (i.e. drought or climate change) and their exacerbation of preexisting grievances and conflicts (Olaniyan and Okeke-Uzodike 2015, Lenshie *et al* 2020).

In 25% of the articles, the stated objective was to analyze conflict-management strategies of people affected by farmer–herder conflicts (Eke 2020, Mbih 2020, Yakubu *et al* 2021). Two examples include Ahmed and Kuusaana (2021), who studied the viability of ranches to prevent farmer–herder conflicts, and Turner *et al* (2011), who studied four communities in Niger to determine which peacebuilding strategies effectively reduced conflicts. About 15% of the articles studied the impact of conflict on the livelihoods of farmers, herders, and other community members (Turner *et al* 2011, Adelakun *et al* 2015, Mwikali and Wafula 2019). For example, in one study a high risk of conflict led farmers to minimize their time in the fields, which decreased their agricultural output (Yusufu *et al* 2020).

The 53 studies were conducted in nine countries, the majority of which were in West Africa (figure 2). Each focused on a single country of interest, suggesting a lack of comparative empirical cases in the literature. Of the reviewed articles, 47% were conducted in Nigeria, 25% in Ghana, 9% in Tanzania, and 6% in Mali. Only a few studies were conducted in Cameroon, Niger, Kenya, Senegal, and Sudan. Figure 2 summarizes the country distribution of the studies. Figure 3 shows the number of articles published per year since 2000, beginning with very low

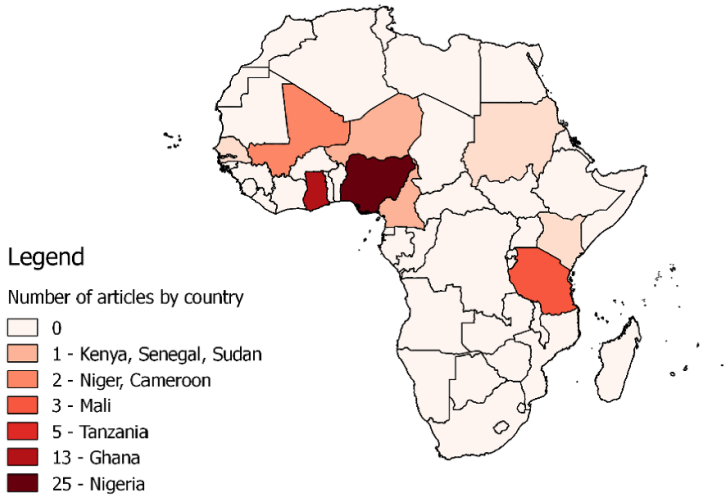

Figure 2. Country distribution of case studies reviewed.

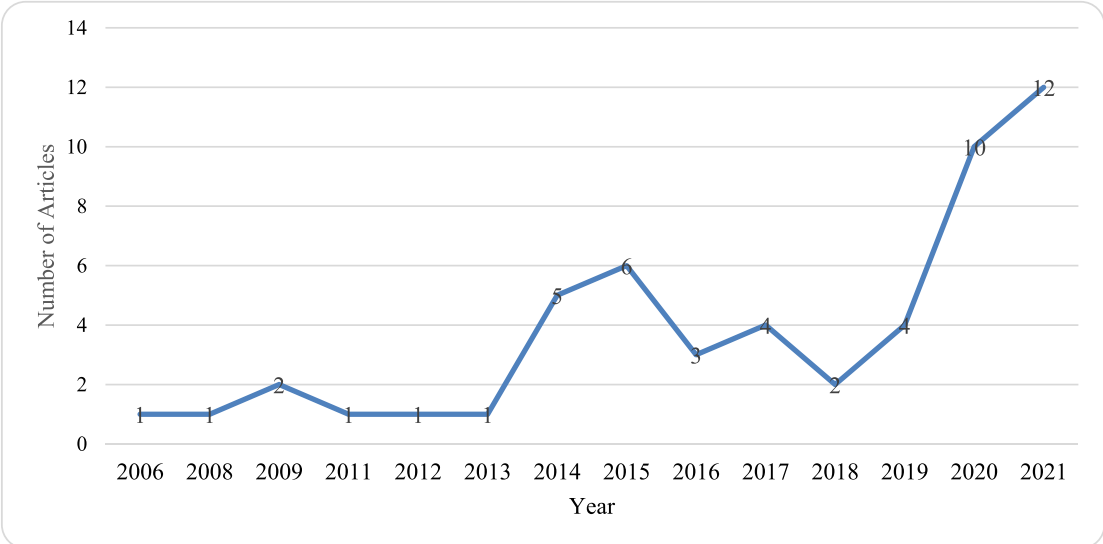

Figure 3. Qualifying academic publications per year (2000–2021).

numbers in the early 2000s and picking up significantly in late 2020 and 2021.

3.2. Main themes

The reviewed literature covered 13 main themes, including conflict, political ecology, agriculture, and climate change. Other themes included conflict management, governance, land tenure, media discourse, and local peacebuilding. Since the studies selected for this review were published across numerous disciplines, these themes reflect the diverse approaches taken by authors of these studies rather than anything inherent to the issue itself we acknowledge that the frameworks and methods used by different authors, whether qualitative, quantitative, or mixed, ultimately shaped their findings and therefore our conclusions. Figure 4 summarizes the key themes identified in the articles. In the following sections we discuss the themes that appeared most frequently, treating climate, migration, and security as one nexus theme due to their interconnectedness.

3.2.1. Conflict

Conflict was a dominant theme in 85% of articles, most of which focused on the root causes and dynamics of conflict (e.g. Benjaminsen *et al* 2009, Akinyemi and Olaniyan 2017, Benjaminsen and Ba 2021) rather than on conflict management (Sangotegbe *et al* 2016, Hamisu *et al* 2017, Paalo 2021) or peacebuilding strategies (Bukari *et al* 2018, Setrana and Kyei 2021). Authors conceptualized conflicts slightly differently. For example, some authors distinguished between direct and structural, or latent, conflicts. Brottem (2016) and Feldt *et al* (2020) conceptualized latent conflict as a constant tension in farmer–herder relationships, while direct conflict

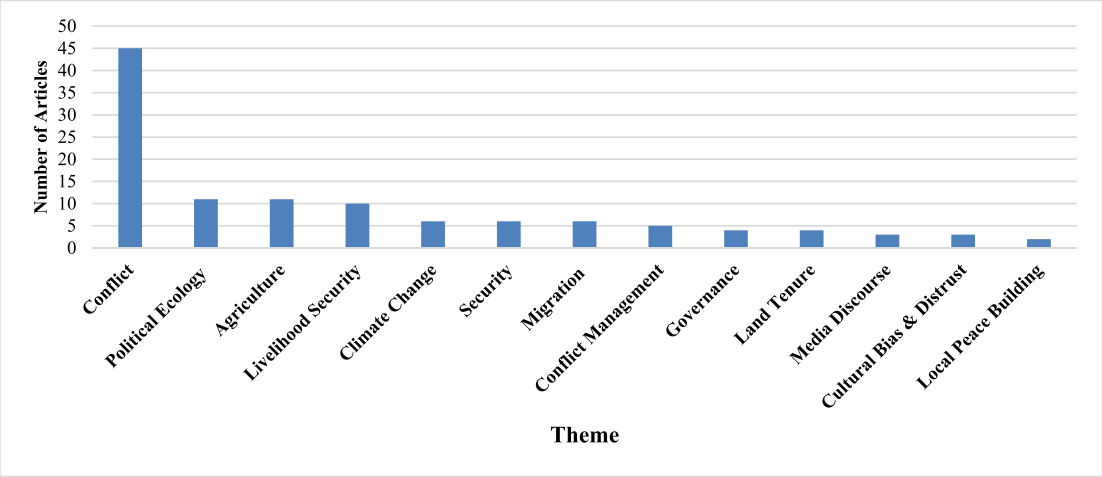

Figure 4. Summary of themes covered in articles.

was triggered by specific incidents that often resulted in violence. Similarly, Kuusaana and Bukari (2015) differentiated between acute and historical conflicts, the latter described as being 'intertwined with issues of belonging and citizenship' with deep historical roots (p. 54). Others focused on overt conflict by examining the most salient 'triggers' that sparked violent clashes (Bwala and Abdulazeez 2020), including farmers secretly killing cows or herders not reporting crop damage by their cows. Others noted conflict as being intertwined with in-group and out-group identities, showing, for example, that historically rooted stereotypes, whereby herders were labeled as aggressors and 'savage nomads' and farmers as nonviolent victims, triggered conflicts (Eke 2020).

3.2.2. Political ecology

Political ecology was a major thematic element in 21% of the literature. These articles explored the politics of farmer–herder conflicts, especially how power imbalances affected access to land and water and in turn caused conflicts (Barre 2012, Okoli and Atelhe 2014, Ahmed and Kuusaana 2021, Benjaminsen and Boubacar 2021). Because many of the articles were rooted in a political ecology framework, we discuss them in more detail in the theoretical and conceptual framework section below (3.4).

3.2.3. Livelihood security and agriculture

Livelihood security was a major theme in 19% of the articles, with subthemes of food security (Dimelu *et al* 2016, Yakubu *et al* 2021) and effects on the socioeconomic conditions of affected communities (Turner *et al* 2011, Mbah *et al* 2021, Yakubu *et al* 2021). Conflicts created a negative feedback loop that degraded trust between farmers and herders and diminished the socioeconomic security of Fulani herders, whose livelihoods were contingent on access to land, water, and the goodwill of their host community (Dimelu *et al* 2017). Farmers' and herders' activities occurred within the same space and required similar provisions of environmental services. Agricultural issues thus often served as both a cause and an effect of conflicts. For example, in one study, large-scale conversion of land into rice paddies significantly reduced available pastoral land (Benjaminsen and Ba 2021). This, in turn, caused increased competition and conflict, as both farmers and herders ventured into common zones. Other authors found that conflict reduced agricultural output because of destruction of crops by cattle (Dimelu *et al* 2016, Mahmood and Azuaga 2020). This destruction of crops led some farmers to request higher compensation from herders (O'Bannon 2006). Agriculture was also implicated in policy recommendations, especially the recommendation to create 'fodder banks,' or ranches (Nwankwo *et al* 2020, Otu *et al* 2020, Ahmed and Kuusaana 2021, Kugbega and Aboagye 2021), or to implement pastoral and agrarian regulations that mutually regulated both farmers and herders (Mamuda *et al* 2017).

3.2.4. Climate change, migration, and human security nexus

Finally, the nexus of three themes—climate change (Brottem 2016, Akinyemi and Olaniyan 2017, Walwa 2017), security (Akinyemi and Olaniyan 2017, Lenshie *et al* 2020, Otu *et al* 2020, Vahyala 2021), and migration (Semiono and Niboye 2019, Lenshie *et al* 2020, Mbah *et al* 2021)—featured prominently in 33% of the studies. In a study in Kenya, authors claimed that climate change—induced migration reduced the available farming and grazing land, which aggravated other risk factors, increased conflict, and decreased human security (Mwikali and Wafula 2019). In another study climate-induced migration in Nigeria compounded economic insecurity, created distrust between ethnic groups, and led to land-use changes and farmer–herder conflicts (Olaniyan and Okeke-Uzodike 2015). Lenshie *et al*

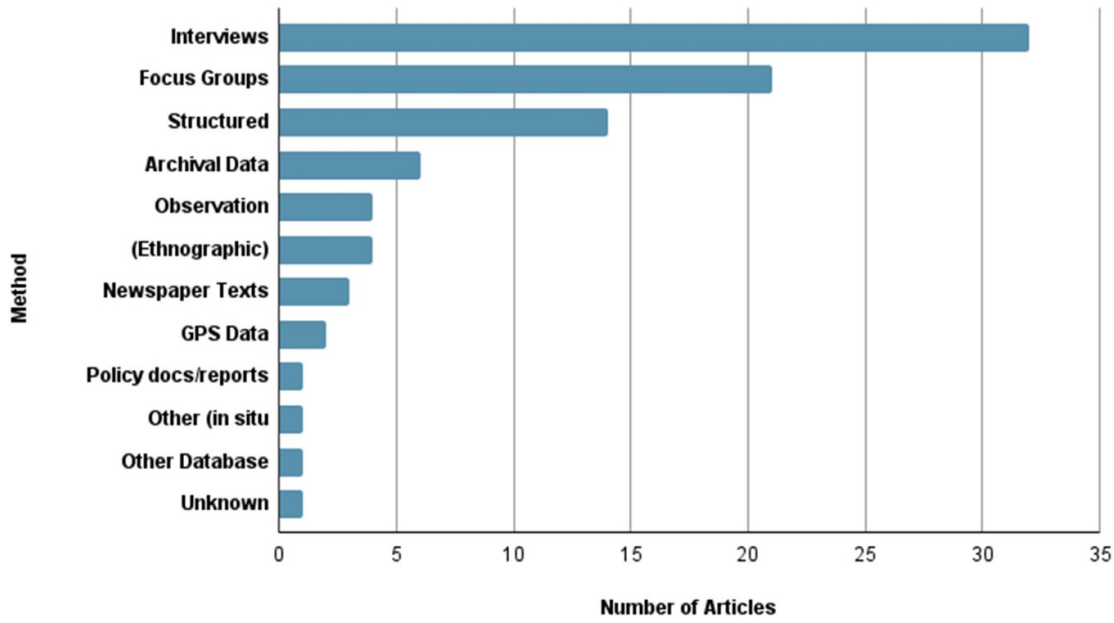

Figure 5. Summary of data-collection techniques used.

(2020) found that droughts in Nigeria were causing migration into 'ungoverned' space. This in turn, they argued, was linked to an increase in violence because of a proliferation of weapons and increased tension between migrating people and local communities. Brottem (2016) described how herders migrated quickly through the central region, causing lots of damage and conflict, and reached the south before farmers had completed their annual harvests, thus creating more damage and conflicts. This kind of crop destruction by migrating herders' cattle was widely reported in the literature as an immediate trigger for farmer–herder conflicts. Farmers responded to the damage with retributive violence, such as killing cattle and injuring or killing herders.

3.3. Methods and analytical techniques

The greatest proportion (58%) of reviewed articles used only qualitative research methods (Walwa 2017, Otu *et al* 2020, Ahmed and Kuusaana 2021). Approximately 25% of articles used mixed methods to gather both qualitative and quantitative data (Turner *et al* 2011, Adam *et al* 2015, Brottem 2016). For example, Feldt *et al* (2020) drew from GPS data on cattle movement and distribution, a land-use classification database, and map-based interviews with herders and farmers. Finally, 17% of the articles utilized purely quantitative methods (Sangotegbe *et al* 2016, George *et al* 2021, Yakubu *et al* 2021), using data collected from structured questionnaires (Hamisu *et al* 2017) or databases (George *et al* 2021).

Most articles in this review relied on a combination of data-collection methods, the top three being interviews, focus group discussions (FGDs), and structured questionnaires or surveys (Akinyemi and Olaniyan 2017, Bukari *et al* 2018, Mahmood and Azuaga 2020). Figure 5 illustrates the number of articles that used each method. Face-to-face interviews with key informants and FGDs were the most common techniques, with 60% of the articles using these methods (Opoku 2014, Kuusaana and Bukari 2015, Tade and Yikwabs 2019). Four groups of authors adopted an ethnographic approach (Olaniyan *et al* 2015, Bukari *et al* 2018, Otu *et al* 2020, Setrana and Kyei 2021) by pairing interviews, FGDs, and surveys with field observations. Some studies used data from newspapers (Chiluwa and Chiluwa 2022, Nwankwo *et al* 2020), GPS or other geographic datasets (Brottem 2016, Feldt *et al* 2020), and archives (Benjaminsen and Boubacar 2021, Nwozor *et al* 2021, Paalo 2021).

The great majority of the articles used some form of qualitative data analysis, including thematic analysis (Tade and Yikwabs 2019, Setrana and Kyei 2021), content analysis (Opoku 2014, Semiono and Niboye 2019, Kugbega and Aboagye 2021), descriptive narrative analysis (Kugbega and Aboagye 2021, Nwozor *et al* 2021), inductive/deductive theoretical analysis, and comparative analysis (Bukari *et al* 2018, Nwankwo *et al* 2020). Articles with mixed or quantitative methods used a variety of statistical techniques to analyze quantitative data such as rainfall levels, incidents and spatial distribution of violence, survey responses, and socioeconomic indicators. The data analysis methods are summarized in figure 6.

3.4. Theoretical and conceptual frameworks

The articles utilized 13 theoretical frameworks across different disciplines. These frameworks are summarized in table 2. Though there may be overlap

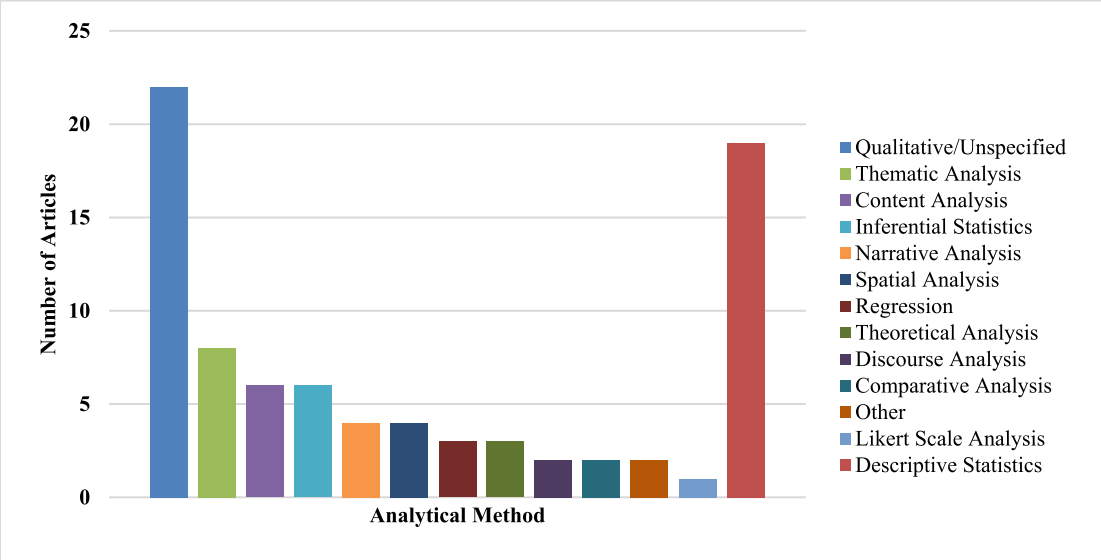

Figure 6. Data analytical techniques.

Table 2. Summary of theoretical frameworks in reviewed articles.

Theories	Count	%	Representative articles
Not specified	18	34%	Kuusaana and Bukari (2015), Mwamfupe (2015), Bwala and Abdulazeez (2020), Feldt *et al* (2020), Mahmood and Azuaga (2020), Arowosegbe (2021)
Political ecology	11	21%	Brottem (2016), Semiono and Niboye (2019), Mbih (2020), Benjaminsen and Ba (2021)
Resource access, scarcity, competition	10	19%	Semiono and Niboye (2019), Otu *et al* (2020), Yusufu *et al* (2020), Paalo (2021)
Rational choice	4	8%	Barre (2012), George *et al* (2021), Nwozor *et al* (2021), Vahyala (2021)
Climate-conflict nexus	2	4%	Okoli and Atelhe (2014), Akinyemi and Olaniyan (2017)
Ungoverned space	2	4%	Mamuda *et al* (2018), Lenshie *et al* (2020)
Conflict framing	2	4%	Chiluwa and Chiluwa (2022), Eke (2020)
Structural violence	2	4%	Dimelu *et al* (2016), Mwikali and Wafula (2019)
Local peacebuilding	2	4%	Bukari *et al* (2018), Setrana and Kyei (2021)
Meta-conflict	1	2%	Adam *et al* (2015)
Social constructivist	1	2%	Kugbega and Aboagye (2021)
Post-structuralist discourse	1	2%	Nwankwo *et al* (2020)
Frustration-aggression	1	2%	Mbah *et al* (2021)
Cognitive appraisal	1	2%	Adam *et al* (2015)
Other	2	4%	Yembilah and Grant 2014, Tade and Yikwabs (2019)

between some categories, we used our discretion to organize the theoretical frameworks into dominant categories. In the largest group of articles (34%), authors did not explicitly name the framework that guided their analysis. Authors of the next largest group of articles (21%) identified political ecology as their theoretical framework for analysis (Turner *et al* 2011, Brottem 2016, Mbih 2020, Benjaminsen and Boubacar 2021). These articles investigated the political power and environmental dynamics in which conflicts were embedded. For example, Ahmed and Kuusaana (2021) situated their study of cattle ranches in Ghana within a political ecology framework to tease out the political, material, and geographic dimensions of farmer–herder conflicts over land and water. In 19% of the studies, the authors employed various theoretical frameworks related to resource access, scarcity, and competition, including land competition theory (Alhassan 2017), relative deprivation theory (Olaniyan *et al* 2015, Akinyemi and Olaniyan 2017), resource scarcity theory (Semiono and Niboye 2019), environmental scarcity theory (Dimelu *et al* 2016, Paalo 2021), and resource access theory (Turner *et al* 2011, Walwa 2020, Yusufu *et al* 2020).

Rational choice theory was used in 8% of articles. Articles that drew on this theory argued that both herders and farmers acted rationally to optimize the individual and collective outcomes of their choices. In short, any actor who used violence did so

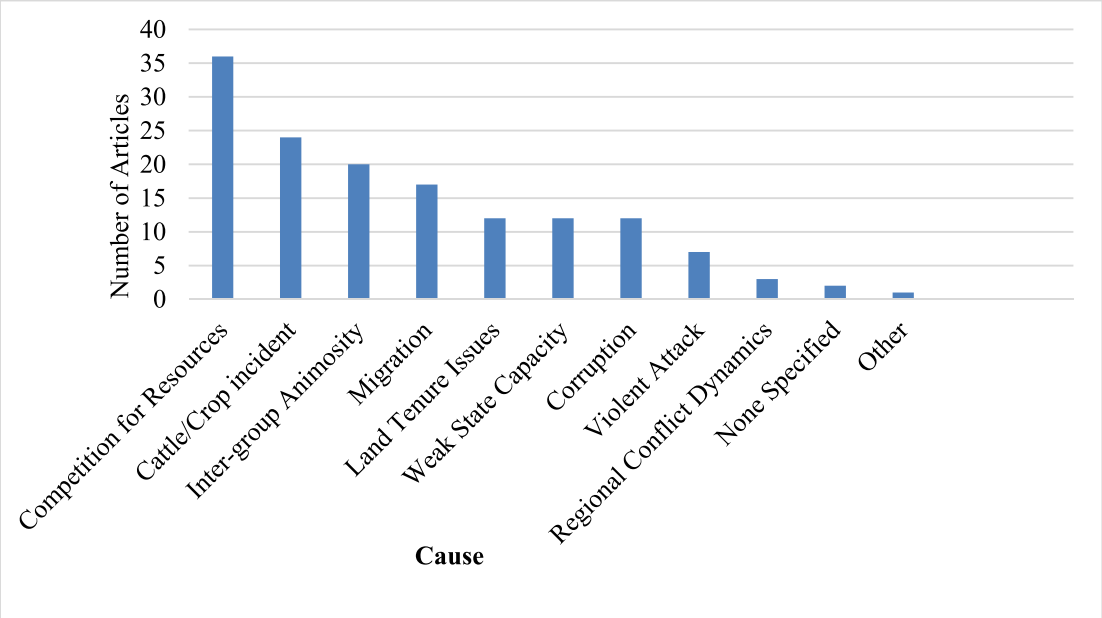

Figure 7. Summary of major causes identified in the review.

because the potential benefits outweighed the costs. Hence, when actors drew immediate or remote benefits from recurrent or protracted conflicts, they were less inclined to resolve fundamental problems. In one example, the authors examined farmer–herder conflicts through the lens of security management and suggested that sanctions (punishment) played a critical role in altering the cost-benefit analysis of conflict actors (Vahyala 2021).

A small portion of the articles (4%) explicitly used a climate-conflict nexus as a framework (Okoli and Atelhe 2014, Akinyemi and Olaniyan 2017). Although many authors noted the influence of climate change on conflict dynamics (Semiono and Niboye 2019, Ahmed and Kuusaana 2021), they eschewed a framework based on the climate-conflict nexus in favor of a more complex causal relationship. For example, Walwa (2020, p. 366) asserted that farmer–herder conflicts 'need to be understood in terms of the power relations (bundles of power) through which resources are negotiated.' Similarly, Brottem (2016, p. 548) argued that the environmental dimensions of conflict are geographically specific and influenced by specific political and social factors.

Only two articles explicitly adopted a peacebuilding framework. One of these drew on local peacebuilding theory to identify ways in which both local and diaspora actors contributed to peace interventions in Agogo in the Ashanti region of Ghana (Setrana and Kyei 2021). Five articles drew on conflict theories, including structural violence (Dimelu *et al* 2016, Mwikali and Wafula 2019), meta-conflict (Adam *et al* 2015), and conflict framing (Chiluwa and Chiluwa 2022, Eke 2020). The authors using conflict framing studied the representation of herders as 'terrorists' and 'jihadists' by media outlets and the implications of these representations for conflict (Chiluwa and Chiluwa 2022, Eke 2020).

3.5. Major causes of farmer herder conflicts

The review revealed eight major causes of farmer–herder conflict (see figure 7), as reported in the reviewed studies. While other causal factors may contribute to the conflicts, the causes highlighted in this review reflect those identified in the reviewed articles. Additionally, because the causes reflect the perceptions of farmers and herders who served as study informants, they sometimes diverge. For example, farmers often assigned blame to herders and vice versa (Opoku 2014, Bwala and Abdulazeez 2020, Benjaminsen and Boubacar 2021). In 68% of the articles, authors identified the major cause of conflict as resource competition, especially competition for arable land, grazing land, and water access (Mamuda *et al* 2017, Feldt *et al* 2020, Lenshie *et al* 2020). Within this causal category, population growth (Okoli and Atelhe 2014, Feldt *et al* 2020, Nwozor *et al* 2021) and changes in land use, such as the conversion of land for large-scale agriculture, led to conflict by exacerbating competition (Sangotegbe *et al* 2016, Alhassan 2017, Walwa 2020). The second most common cause of conflict, cited in 24% of articles, was damage involving either cattle or crops. These incidents included cattle trampling or eating crops, farmers or other locals rustling cattle, farmers blocking cattle migration paths, and cattle polluting water points (Brottem 2016, Otu *et al* 2020, Yakubu *et al* 2021). In the study by Kuusaana and Bukari (2015), for example, crop destruction was identified as an immediate trigger for farmer–herder clashes in the

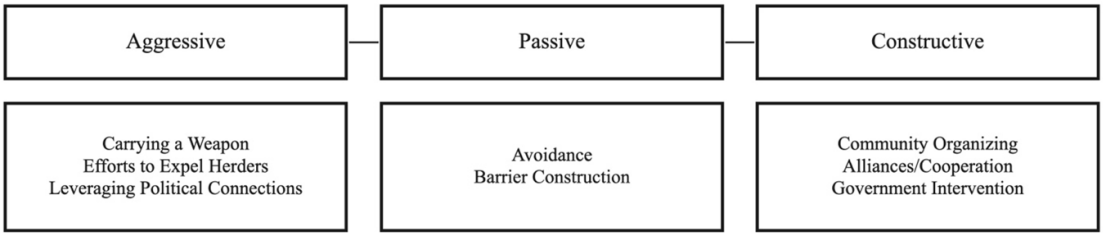

Figure 8. Strategies to control and prevent conflicts.

Agogo area, although other latent causes may have also factored into the conflicts.

Intergroup animosity—defined here as negative perceptions of or hostilities towards people from a different social, religious, or cultural group—was the third most identified cause of conflict, appearing in 20% of the articles (Turner *et al* 2011, Mahmood and Azuaga 2020, Ahmed and Kuusaana 2021). For example, Mahmood and Azuaga (2020) examined the biases of farmers and herders and concluded that both groups perceived their own identity and livelihood as superior. These perceptions fed into and escalated conflicts over scarce resources. In another study, Alhassan (2017) demonstrated that chiefs and landowners in Kwahu, Ghana, viewed ethnic Ewe farmers as 'settlers' despite their having lived in the area for generations. Ewe farmers, in turn, viewed Fulani as 'aliens' even though many held certificates of residence. Similarly, Ahmed and Kuusaana (2021) showed that social cleavages along ethnic lines and other perceived differences factored into farmer–herder conflicts.

The fourth most common cause of conflict was migration (Lenshie *et al* 2020, Ahmed and Kuusaana 2021). One interesting case, presented by Olaniyan and Okeke-Uzodike (2015), demonstrated how an increase in climate-induced migration of one Fulani group (Bororo) sparked violence between a settled Fulani group (Ibile) and Yoruba residents in Nigeria, heightening ethnic differences and tension where the groups had previously coexisted. Historical patterns in migration also factored into current conflicts. Mwikali and Wafula (2019) discussed how the British colonial government evicted pastoralists from forested land in Kenya to conserve it. Later, the Kenyan government upheld these policies that excluded herders from traditional grazing lands, forcing pastoralists to search for pasture in places that brought them into conflict with other residents. In another article, Mwamfupe (2015) explained how, in the past, the Tanzanian government had leased land to private hunters, pushing Maasai pastoralists to migrate to find new grazing land. In the process, these pastoralists encroached on agricultural land. This finding demonstrates how groups who identify as victims of displacement may be viewed by other groups as perpetrators.

Another related structural cause of conflict was land tenure insecurity. Some authors found that ambiguous or outdated land laws have led farmers and herders to contest land rights (Kuusaana and Bukari 2015, Dimelu *et al* 2016, Mamuda *et al* 2017). In another study most farmers possessed land titles, yet herders obtained grazing permits from local authorities to access farmers' land (Feldt *et al* 2020). Similarly, Kuusaana and Bukari (2015) reported that up to 70% of the land in the study area fell under the control of chiefs who granted customary rights to Fulani herders under both formal and informal arrangements. Migratory herders sometimes stayed year-round or acquired additional land from smallholders or chiefs. Furthermore, local authorities also revoked land rights of farmers and herders to lease land to private agribusinesses. This contested and insecure land tenure created tensions within the local population and gave farmers a sense of land insecurity. Other studies linked state weakness or ineffective policy responses to farmer–herder conflicts, especially in Nigeria (Lenshie *et al* 2020, Nwozor *et al* 2021), where people viewed the Nigerian government's lack of action against herders who illegally possessed weapons as a sign of government support (Mamuda *et al* 2018). Other scholars pointed to a more intentional link between state or nonstate armed groups and herders (Mbah *et al* 2021). For example, affiliates of Boko Haram attacked herders and destroyed grazing land, which led some herders to arm themselves for self-defense (George *et al* 2021).

3.6. Strategies for the prevention of farmer–herder conflicts

Farmers and herders have adopted diverse strategies that they believe can prevent conflict outbreaks and escalating violence. We have categorized these strategies along a spectrum from aggressive (e.g. carrying weapons), through passive (e.g. avoidance of confrontation), to constructive (e.g. community organizing), altogether illustrated in figure 8 with examples.

3.6.1. Aggressive

The studies found that farmers and herders armed themselves to deter confrontations or defend themselves (Benjaminsen *et al* 2009, Chiluwa and Chiluwa 2022, Eke 2020). Authors reported that herders and ethno-religious groups in the Sahel sometimes carried weapons to protect themselves and their cattle. However, rather than preventing or controlling conflict, unrestricted access to weapons may be a conflict driver. In one study, 30% of the respondents resorted to 'self-defense' or 'self-help' strategies that sometimes led to violent outcomes (Akinyemi and Olaniyan 2017). The key informants in several studies claimed that the proliferation of small arms in their region escalated farmer–herder conflicts (Lenshie *et al* 2020, Nwankwo *et al* 2020, Nwozor *et al* 2021). In one study, herders carried weapons to defend cattle from 'criminals,' yet other informants perceived herder violence as indiscriminate and tacitly supported by the Nigerian State (Mbah *et al* 2021). Brottem (2016) pointed out how the lack of trust among people made it easy for misperceptions to degenerate into violence.

3.6.2. Passive

Farmers and herders used avoidance strategies to prevent or control conflicts (Adam *et al* 2015, Adelakun *et al* 2015, Mwikali and Wafula 2019, Tade and Yikwabs 2019). Both made livelihood-related transitions due to pasture and cropland shortages, labor emigration, and shifts in livestock ownership. As cropland diminished, herders moved farther from farming areas to graze their animals during the cropping season, only returning after harvest. They used designated stock routes to prevent crop damage (Bwala and Abdulazeez 2020) and split herds into smaller sizes to avoid fields (Adam *et al* 2015). They avoided towns when they feared violent reprisals from the local community (Alhassan 2017). 'Fodder banks,' a preventative policy solution, was proposed by a local district assembly in Ghana but was not implemented because it required land and greater support by local leaders (Alhassan 2017). Some cattle owners also engaged livestock-rearing specialists, who diligently managed cattle outside of the village territories and within the village to avoid crop destruction (Turner *et al* 2011). Farmers avoided conflict by not going to their fields when cattle herds were nearby, changing crops (e.g. not growing plantains, which require 12 months to mature), and changing growing schedules (Opoku 2014, Kugbega and Aboagye 2021). Female farmers did not go to their farms when herders were spotted grazing around them for fear of being sexually assaulted or killed (Olaniyan *et al* 2015). Some farmers decided to reduce their farming and engage in supplemental off-farm income-generating activities (Yikwab and Tade 2021).

3.6.3. Constructive

Constructive strategies to avoid conflict included community organizing and other activities that developed inter-group relationships (Ofuoku 2009, Kuusaana and Bukari 2015, Walwa 2017). These activities typically engaged community stakeholders through education campaigns or dialogue platforms. Many of the articles recommended awareness-raising and educational campaigns as a policy solution to farmer–herder conflicts (Adelakun *et al* 2015, Bwala and Abdulazeez 2020, Ahmed and Kuusaana 2021). In a study in northwest Cameroon, dialogue platforms were found to be successful mechanisms for promoting cooperation and direct problem-solving between farmers and herders (Mbih 2020). The facilitators were from local communities and had deep knowledge of the agro-pastoral issues that contributed to conflicts. This initiative helped address conflicts before they became violent. Some studies found that dialogue was not effective when one group participated more in these activities than the other (Kuusaana and Bukari 2015, Feldt *et al* 2020). Several articles found that 'alliance farming', which, like cooperatives, involves farmers organizing into groups to pursue specific benefits such as low interest loans for equipment, helped to prevent conflict (O'Bannon 2006, Bukari *et al* 2018, Feldt *et al* 2020). For example, the participation of farmers and herders in alliance farming promoted more peaceful coexistence and conflict resolution in Northwest Cameroon by increasing dialogue and social interaction (Feldt *et al* 2020, Mbih 2020). Similarly in Ghana, informal partnerships with one another for mutual benefit generally promoted peaceful coexistence among farmers and herders (Yembilah and Grant 2014a).

3.7. Conflict management and peacebuilding: approaches, principal actors, and their roles

Conflict management and peacebuilding approaches are those that are deployed once conflicts have erupted. We identified three principal approaches to farmer–herder conflict management in our review: third-party intervention, litigation and adjudication, compensation, and police involvement. Below we analyze each of these approaches.

3.7.1. Third-party intervention

In 55% of the articles, third-party intervention was the most common approach for addressing farmer–herder conflicts (Benjaminsen *et al* 2009, Adelakun *et al* 2015, Brottem 2016, Mwikali and Wafula 2019, Mbih 2020). In most of these cases, conflicts were addressed at the community level, often with leaders arbitrating between conflict parties (Kugbega and Aboagye 2021, Paalo 2021, Vahyala 2021). These interventions included mediation, arbitration, and informal consultation. In Senegal, for example, local

leaders facilitated decision-making about cattle corridors and migration timelines (O'Bannon 2006). The resulting agreements placed responsibility on both farmers and herders: farmers had to monitor their fields every day and herders had to accompany their cattle and confine them at night. Bukari *et al* (2018, pp. 89–90) examined variations in violent conflict between the regions of Ghana. They found that the relative lack of violence in Kowereso could be explained, in part, by the interventions of elected community members who served on 'unit committees.' These groups resolved conflicts and de-escalated intergroup tension. In the same article, the authors reported that chiefs determined the amount of compensation cattle owners paid to farmers for damaged crops (Bukari *et al* 2018). In one study, third-party interventions involving elders or village chiefs were found to be highly effective (Dimelu *et al* 2016).

Farmers preferred interventions by village management committees and leaders who could arbitrate and determine compensation. They often did not think that engaging a 'neutral body' or religious leader would lead to effective resolution. In other studies, traditional leaders were critically important players in conflict resolution (Bwala and Abdulazeez 2020). Nearly 40% of studies found that traditional leaders, including religious leaders (Adam *et al* 2015) and village committees (Mwamfupe 2015), managed or resolved conflicts, especially as mediators and consultants (Barre 2012, Alhassan 2017). In one study, ad hoc local government committees in Nigeria were involved in resolving 85% of conflicts. These committees were comprised of state and local governments, police, members of local cattle owners or farmer associations, and the military (Bwala and Abdulazeez 2020). In another, farmers and herders in Borneo State, Nigeria, perceived traditional institutions as credible arbitrators in farmer–herder conflicts because these institutions could 'restore peace' after the conflict. Informants stated that they appreciated the low cost and speed of traditional institutions compared to formal state resolution processes (Hamisu *et al* 2017). In a study on local peacebuilding efforts in the Agogo traditional area, Ghana, both farmers and herders preferred intervention by traditional authorities over the court system (Setrana and Kyei 2021). The chiefs wielded enormous sociopolitical power and could promptly mandate that one party (often the herder) pay compensation to the farmer for damage. Some herders complied because chiefs had the power to revoke their access to land and pasture.

About 25% of the articles noted that community associations were involved in conflict management strategies (Sangotegbe *et al* 2016, Mwikali and Wafula 2019, Benjaminsen and Boubacar 2021). Some of these mediating actors were community farmer associations or community development committees. Community members thought these associations effectively reduced tensions and resolved conflicts. One reason given for their success was that involvement in these associations cut across lines of ethnic and religious groups. People from various ethnic groups had invested in cattle and therefore the conflicts were less defined by the presence of a perceived ethnic 'other' (Ofuoku 2009). In another study, a village environmental committee comprised of farmers and herders resolved conflicts locally but passed difficult cases to higher institutions (Mwamfupe 2015).

Despite the effectiveness of third-party interventions to resolve and manage farmer–herder conflicts, there were limitations to their success. Some studies reported distrust between local agricultural communities and their leaders. Some communities viewed the leaders as corrupt and biased towards herders, which compromised their independence as mediators (Alhassan 2017, Semiono and Niboye 2019, Mbih 2020). In Agogo, Ghana, 85% of the farmers in one study did not trust traditional leaders in matters of conflict resolution (Kugbega and Aboagye 2021), accusing chiefs of engaging in illicit land deals and offering land to herders in exchange for cattle. In a study of conflict in four districts of Tanzania, Mwamfupe (2015) detailed the limited capacity of local leaders and institutions to resolve conflicts due to a lack of training, the dominance of crop farmers in these positions, and a lack of trust by farmers in the process.

3.7.2. Litigation, adjudication, and police involvement

In 30% of the articles, parties to conflicts sought solutions in the law courts (Brottem 2016, Akinyemi and Olaniyan 2017, Benjaminsen and Boubacar 2021, Yakubu *et al* 2021). Yakubu *et al* (2021) found that 99% of their respondents viewed 'arrest and prosecution' as the most effective management strategy. Bwala and Abdulazeez (2020) found a more nuanced attitude, with informants identifying courts as the only actor able to address disputes about grazing land, while other local institutions could settle disputes over water or bush burning. In these studies, some informants reported that they preferred the legal route because it was cheaper (i.e. travel and informal fees) than mediation. They also perceived that the courts were more impartial than local leaders (Opoku 2014, Semiono and Niboye 2019, Nwozor *et al* 2021). Courts also provided a neutral environment in which each side could tell their story (Setrana and Kyei 2021).

The courts, like traditional leader systems, were not immune to perceptions of bias or corruption (Brottem 2016, Walwa 2020, Kugbega and Aboagye 2021). Informants complained that the legal system was a 'cash cow' for the government because citizens had to bribe judges and pay exorbitant court fees (Benjaminsen and Boubacar 2021). Although farmers sought solutions through the court system, they perceived it to be influenced by wealthy cattle

owners (Semiono and Niboye 2019, Kugbega and Aboagye 2021). Similarly, Brottem (2016) found that cattle owners (as distinct from herders who watched their cattle) bribed court officials. Benjaminsen and Boubacar (2021, p. 440) found that fewer farmers pressed charges after realizing that pastoralists could bribe police and court officials. Walwa (2020) reported that farmers, more than herders, used the courts to settle disputes, but that farmers complained that herders paid bribes to win court cases, though judicial officials denied this. In that study, farmers did routinely lose cases in court because they either lacked concrete evidence to support their cases or presented only oral evidence or they lacked evidence from the agricultural officers on whose accounts of the cases the courts may have relied heavily on. In some studies, informants did not choose the court system because it consumed time and money and interfered with livelihood activities (Brottem 2016, Semiono and Niboye 2019, Vahyala 2021). Where the court system was used, punishment did not always deter violence, and the slow legal process even led some people to 'settle scores locally,' in other words, retaliate directly rather than seeking legal solutions (Vahyala 2021). Therefore, some victims only used the courts as a last resort (Benjaminsen and Ba 2009).

Law enforcement agencies (primarily the police) were also involved in conflict resolution, sometimes in collaboration with the courts and other times as independent mediation actors. Police were involved in conflict management in 13% of the articles. They investigated incidents of crop destruction (Adelakun *et al* 2015, Kugbega and Aboagye 2021) or ensured that conflicting groups paid compensation for damage (Benjaminsen *et al* 2009, Bukari *et al* 2018). The effectiveness of the police, however, was mixed. While some informants viewed intervention by law enforcement agents as constructive (Adelakun *et al* 2015), others found them to be largely ineffective (Mwamfupe 2015, Bwala and Abdulazeez 2020). One common criticism was that cattle owners bribed the local police to return their confiscated cattle. This real or perceived corruption eroded the trust that people had in the authorities and their ability to be neutral interveners (Benjaminsen *et al* 2009). Informants in other studies said that police were outsiders to the situation (Setrana and Kyei 2021), favored herders (Yembilah and Grant 2014b), took both cash and in-kind bribes (Benjaminsen *et al* 2009, Vahyala 2021), were slow to respond to conflicts (Kugbega and Aboagye 2021), and often failed to reconcile conflict parties (Setrana and Kyei 2021).

3.7.3. Compensation and direct farmer–herder negotiations

The third most common conflict management approach used by farmers and herders was to negotiate and pay compensation for damages to crops, cattle, or fencing. Payment usually took the form of either money or bags of fertilizer (Ofuoku 2009, Dimelu *et al* 2016, Alhassan 2017, Yakubu *et al* 2021). Sometimes, an authority determined the amount, but in other cases, conflict parties negotiated it themselves (Barre 2012, Adam *et al* 2015, Mwamfupe 2015). In 19% of the articles, farmers and herders were the primary actors involved in conflict resolution, that is they negotiated the resolution process without involving any external actors. Although compensation was a conflict-resolution strategy, it generally failed to resolve the conflict or address the underlying factors. One study found that farmers demanded more from herders when their agricultural yields were low (O'Bannon 2006). Requests for compensation could escalate conflicts when the requested amount was too high (Bukari *et al* 2018). Some herders accused farmers of destroying their own crops to extort payment from them (Alhassan 2017). Farmers sometimes accused herders of bribing local officials to avoid paying (Benjaminsen *et al* 2009). Some farmers never received the agreed-upon compensation and grew more resentful (Brottem 2016).

3.8. Limitations

There are two main limitations to this review worth noting. First, given that this is a review paper rather than one based on primary research, the findings and conclusions presented below reflect the nature of the evidence presented in the reviewed articles, which may have their own weaknesses and biases from the informants and respondents recruited by the different authors. Second, while our review yielded valuable insights, exclusively using peer reviewed articles may have led to the omission of significant insights and themes from publications outside journals, including policy reports, media sources, and government documents.

4. Discussion and conclusion

Despite a relatively large body of literature on farmer–herder conflicts in sub-Saharan Africa, where these conflicts have grown extensively in the past few decades and pose a security threat, no studies have synthesized the historical and current state of knowledge of these conflicts. We reviewed and synthesized 53 empirical studies on farmer–herder conflicts in sub-Saharan Africa to explore the drivers and impacts of the conflicts and the opportunities and limitations of resolution and peacebuilding strategies. This synthesis of knowledge advances several strands of literature on farmer–herder conflicts, including geography, peace and conflict studies, natural resources and the environment, and African studies.

Each of the studies we reviewed focused on conflicts in a single country, with the great majority looking at one of nine countries (Nigeria, Ghana, Tanzania, Mali, Cameroon, Niger, Kenya, Senegal, and Sudan). These countries fall within the Savannah

and Sahelian regions, where animal rearing is both a major livelihood and an important cultural identity. They are also endowed with rich forage and water resources to support animal rearing. The literature confirmed that environmental factors have induced seasonal migration of herders across expanded regions in the search for water and fodder for cattle. In particular, many herders have moved with their cattle across the Sahara into transitional forest zones and further into rainforests for better forage, putting pressure on these lands. During this migration, they have negotiated with local landowners to traverse their lands in return for various benefits. Such arrangements have sometimes been facilitated by powerful local middlemen, who receive both immediate and remote benefits. Middlemen brokered between traditional leaders and herders.

Competition for resources (agricultural land, grazing land, and water) remains the primary driver of farmer–herder conflict, followed by direct crop damage by cattle, and preexisting intergroup grievances. Yet it was also clear that complex interactions between sociocultural, political, and environmental factors both underlay and exacerbated the conflicts. For example, climate change has contributed to changes in the migratory patterns of herders, which has compounded ethnic discrimination (i.e. farmers perceiving all herders as foreign migrants), poverty, and land tenure insecurity. North Africa and the Sahelian regions have witnessed the mass movement of herders into more resource rich regions because of climate change. However, local smallholders have resisted these movements, opposing the intrusion of herders into their farmlands and active fallow lands. In some cases, farmer–herder conflicts have been influenced by external factors such as militia groups. A few studies mentioned the influence of external armed groups on farmer–herder conflicts (Benjaminsen and Boubacar 2021, George *et al* 2021).

The customary land tenure system practiced by many countries in sub-Saharan Africa has also created further complications in landholding, land use, and land transfer. For example, smallholders often establish their farms in grazing corridors or along bodies of water where cattle drink. The resultant growth in contestations over land ownership and boundaries between farmers and herders has exacerbated issues of land tenure insecurity. The literature also described instances in which herders were accused of driving their cattle into active or fallow farmlands to access grazing lands and water resources. This combination of processes has led to increased intergroup tension and, in some cases, violent conflicts.

Increasing commercialization and large-scale investment in land for agricultural purposes was also cited as a cause of the rise in farmer–herder conflicts. Since 2008, the global food, fuel, and financial crises have driven many countries in Europe, Asia, and America to acquire large lands in Africa for investment in food, fodder, biofuel, and forestry. These large commercial agricultural holdings reduce the amount of land available to smallholder farmers, which ultimately increases competition between farmers and herders on the remaining land. As crop destruction occurred repeatedly, farmer distrust of herders due to crop destruction deepen over time and create volatile conditions for conflicts to escalate. Thus, agriculture can be understood as both a driver (in terms of growing large scale commercial agriculture) and casualty (in terms of smallholder farmers becoming victims of crop damage) of farmer–herder conflicts.

Cultural and ethnic factors that drove farmer–herder conflicts usually had a historical foundation and were tied to the lived experiences of both farmers and herders in host communities. Herders tended to hold their cattle in higher esteem than crops, and farmers held the opposite opinion. Farmers often perceived that they were disadvantaged by the mere presence of herders and the systemic support herders received from chiefs and other powerful local elites. They accused herders of deliberately allowing their cattle to enter farms and feed on crops. On the other hand, herders accused farmers of exaggerating minor damages to their farms and crops. Thus, identity constructions threatened peaceful relations between farmers and herders and limited opportunities for peacebuilding. In some instances, because both farmers and herders considered their livelihoods to be superior, they failed to appreciate the value of the others' investments.

In reviewing these articles, we noted inconsistencies in the use of terminology that further exemplified these identity-based tensions. Authors identified farmers as smallholders (Kuusaana and Bukari 2015), settlers (Alhassan 2017, Kugbega and Aboagye 2021), or indigenes (Barre 2012, Yembilah and Grant 2014b) in Ghana or by their ethnicities, such as Dogon in Mali (Benjaminsen and Boubacar 2021) or Ewe (as opposed to the Kwahu chiefs) in Ghana (Alhassan 2017). Herders were described as Fulani (Bukari *et al* 2018, Benjaminsen and Boubacar 2021, Setrana and Kyei 2021), semi-nomadic herders (Brottem 2016), or pastoralists (Kugbega and Aboagye 2021). One article described both groups as 'victims' (Tade and Yikwabs 2019). Most articles treated farmers and herders as discrete groups, although a few authors noted the ambiguity and overlap of identity categories (Turner *et al* 2011). For example, some farmers owned cattle (George *et al* 2021) and some herders held certificates of residence (Alhassan 2017) or were nonmigratory (Kugbega and Aboagye 2021). Some authors' language revealed bias against herders or Fulani. For example, Otu *et al* (2020) referred to herders as 'the menace.'

We found that various tactics were used to prevent potential conflict, including those that could be categorized as aggressive, passive, or constructive.

Both farmers and herders used aggressive conflict prevention strategies through the wielding of arms or machetes for self-defense. Yet while farmers accused herders of using weapons to perpetrate heinous crimes, herders often explained that they wielded guns to protect their cattle against localized rustling. Some studies described how farmers formed 'militia' groups to retaliate for damage and fight herder encroachment of land (Benjaminsen and Boubacar 2021, Nwankwo *et al* 2020). Farmers often constructed barriers to keep cattle and herders away from their farms as a preventive mechanism, while herders resorted to isolation from farmed areas where possible to avoid potential confrontation. Herders who were non-indigenous in some instances chose police protection. For farmers, conflict avoidance tactics included simply not going to their farms too early in the morning when herders are more likely to cross their lands with cattle. Some prevention mechanisms were state sponsored, for example the creation of cattle ranches (Ahmed and Kuusaana 2021). Cattle ranching aims to restrict herders from farming activities. However, ranches have yet to be operationalized in many sub-Saharan African countries because they restrict the traditional practice of pastoralism among herders, which relies more on mobility. In the Ghana example (Ahmed and Kuusaana 2021), some herders did not utilize ranches because they perceived that ranches predisposed cattle to diseases because of the limited water and feeding sources and the cohabitation of many cattle from different herds, while others complained that ranches were too distant from community markets and limited the sale of milk and meat. The ranches example underscores the limits to state-centered and top down models of farmer–herder conflict prevention that do not address other underlying locally relevant issues.

Once conflict occurred, farmers and herders responded in different ways. Some farmers changed the crops they grew the following season or their growing schedules, or even reduced farm sizes to minimize potential economic losses and enable them to indulge in additional off-farm employment. Herders, for their part, might reduce their stock sizes (by splitting herds or selling some off), identify new routes (corridors) of movement to avoid croplands, move into barren lands inappropriate for farming, or better define boundaries between farmlands and grazing lands.

Conflict management and peacebuilding efforts involved different actors depending on the scale of the conflicts. In many of the conflicts reported in the literature, third-party interventions were used, particularly mediation, arbitration, and informal consultations. Actors driving these processes included chiefs, assembly members, unit committee members, and farmer unions. In addition, religious leaders, chief butchers, chief cattle dealers, and Muslim sect chiefs were also relevant as third parties. Farmers preferred to refer cases to village management committees for redress. However, herders often accused localized third parties of bias because many of the leaders lived with the farmers in the community and were inclined to side with them. Some also lacked the requisite training to handle the conflict management and peacebuilding processes.

Both farmers and herders also used litigation, compensation, and police involvement to manage and resolve farmer–herder conflicts. The police investigated cases and presented evidence to courts. They also ensured that compensation was paid based on court recommendations. Herders preferred the police because they facilitated access to courts through arrests and prosecution. Farmers preferred the courts for conflict resolution but believed that the police could be bribed. As with local leaders, people also accused local courts and law enforcement agencies of being biased and corrupt (Brottem 2016, Walwa 2020). Farmers accused local judges of accepting bribes from wealthier cattle owners and asking for court fees from poor farmers. Finally, some parties favored compensation in the form of money or bags of fertilizer. These payments were directly negotiated between the farmer and herder. When they could agree on the terms and amount of payment, such cases did not require mediation by third parties or courts. However, farmers demanded more compensation when they anticipated that cattle damage would significantly affect their crop yields and accused local officials of accepting bribes to protect herders from having to pay compensation commensurate with the damage they caused.

These conflict management and peacebuilding approaches had strengths and limitations. No single approach was universally effective across all situations. Notably, lack of trust in local institutions hampered conflict management and peacebuilding. One major criticism in the literature was that conflict parties could bribe third-party actors, especially local authorities and courts, to influence outcomes. This perceived lack of impartiality eroded the conflict parties' trust in institutions and each other. Further, even in cases where these third-party actors did not require bribes, informal fees (i.e. paying for the authority's travel expenses) made nonviolent conflict interventions too expensive for some people. The deteriorating trust between farmers and herders also undermined resolution attempts and created fertile conditions for future conflicts. In some instances, incidents of damage to crops or cattle sparked sudden violent confrontations, which led to heightened tensions and further reduced trust between individuals and groups (farmers and herders). Additionally, some conflict management strategies worked in some contexts but failed in others. For example, while 'fodder banks', or ranches, held promise as a conflict

prevention strategy in some locations (Otu *et al* 2020, Kugbega and Aboagye 2021), elsewhere they fell short because they did not address the challenges posed by climate change, land grabs, insecurity, and the migratory needs of herders and their cattle (Ahmed and Kuusaana 2021). It was clear from the literature that solutions to farmer–herder conflicts may fail if they do not consider the complex ecological, economic, political, and social power dynamics that influence the interactions between ranchers, farmers, and nomadic herders in different conflict contexts.

Overall, nonconfrontational and passive conflict management methods and constructive community-based interventions were the most promising methods for addressing farmer–herder conflicts. Most articles emphasized the importance of community-based organizations in farmer–herder conflict management, while others pointed to ad hoc local government committees as being more effective in conflict resolution in some contexts. Some noted that community stakeholder meetings, educational campaigns, and dialogue platforms were important for conflict resolution. Some evidence thus suggests that such participatory methods of conflict resolution, which places responsibility on both farmers and herders and often involves locally elected committees and traditional institutions through consultations, mediation platforms, and informal agreements have been effective at de-escalating conflicts. However, more studies focusing on the opportunities and limits of such strategies in conflict resolution are clearly needed to establish the extent of their potential impact. More importantly, there our analysis uncovered important gaps in the literature that must be addressed in future work on farmer–herder conflicts to understand the complexities of these conflicts and increase the effectiveness of peacebuilding efforts. Existing studies focus overwhelmingly on drivers of conflicts, with many barely touching peace and conflict resolution methods (Kuusaana and Bukari 2015, Eke 2020, George *et al* 2021). Similarly, while the literature acknowledged that the drivers of farmer–herder conflicts were both latent and direct, most studies focused on the latter and ignored the more latent identify and ethnic-based causes of the conflict. We contend that effective management of these conflicts requires a clear understanding of both the latent and direct causes. There was also a clear overrepresentation of male farmers as informants, as opposed to female farmers, in these studies. However, this would not apply to herders because they tend to be overwhelmingly male. Future work should explore gender dynamics of farmer–herder conflicts, including examining female farmers access to compensation and capacity to engage the police, traditional leaders, and other third parties and conflict management actors in different contexts. Only a third of the studies covered conflict management strategies and peacebuilding processes, suggesting a bias in the literature towards causes. More empirical studies are needed in this area to determine which resolution strategies hold promise for addressing farmer–herder conflicts in different contexts. Further, our analysis revealed a conspicuous lack of comparative case studies on farmer–herder conflicts. All the 53 studies reviewed focused on single countries, most of them in West Africa (47% in Nigeria and 25% in Ghana). Comparative case studies on the topic are needed to better understand the similarities and differences in different contexts. Furthermore, while a significant number of articles did touch on impacts of the conflicts, most were qualitative and cross-sectional in design. More quantitative, longitudinal, and large-N studies with larger samples are needed to document the long-term impacts of farmer–herder conflicts in different contexts on, for example, food security. The literature leaves unaddressed a good understanding of the links, interactions, and feedback loops between different causal factors and outcomes. The intricately complex linkages between factors and causes/effects of these conflicts call for more coupled human-natural systems and agent-based modeling research to gain a more complete understanding of the feedback loops connecting the different factors.

In conclusion, the biggest policy lessons from our review are that farmer–herder conflicts are complex, multifaceted, and shaped by the interplay of several sociopolitical and environmental factors. No state or non-state approaches were independently effective, suggesting that a more effective way to address these conflicts should rely on state and non-state actors working in collaboration. In most cases, policy and institutional weaknesses played a critical role in the both the emergence and resolution of these conflicts. In Nigeria for example, many articles implicated the state as failing to effectively act on farmer–herder conflicts due to its failure to curb illegal arms possession, allowing farmers and herders to possess arms for confrontations and self-defense, which created the right conditions for the conflicts to cascade into violence (Mamuda *et al* 2017, Mbah *et al* 2021).

Data availability statement

The data cannot be made publicly available upon publication because they are owned by a third party and the terms of use prevent public distribution. The data that support the findings of this study are available upon reasonable request from the authors.

Funding

This work was supported by a faculty research Grant from the Kroc Institute for International Peace Studies at the University of Notre Dame.

ORCID iDs

Ellis Adjei Adams https://orcid.org/0000-0003-3783-9005
Anna Mittag https://orcid.org/0000-0003-0346-1307

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